Publication Clusters

COM-480 · Data Visualization · Milestone 2

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Link to the website skeleton: https://com-480-data-visualization.github.io/publication-clusters/

Project Goal

We visualize the evolution of global academic influence by mapping citation networks between universities on an interactive 3D globe. Citations are treated as directed geographic vectors, allowing us to track how the flow of scientific knowledge shifts across decades.

The tool answers three questions:

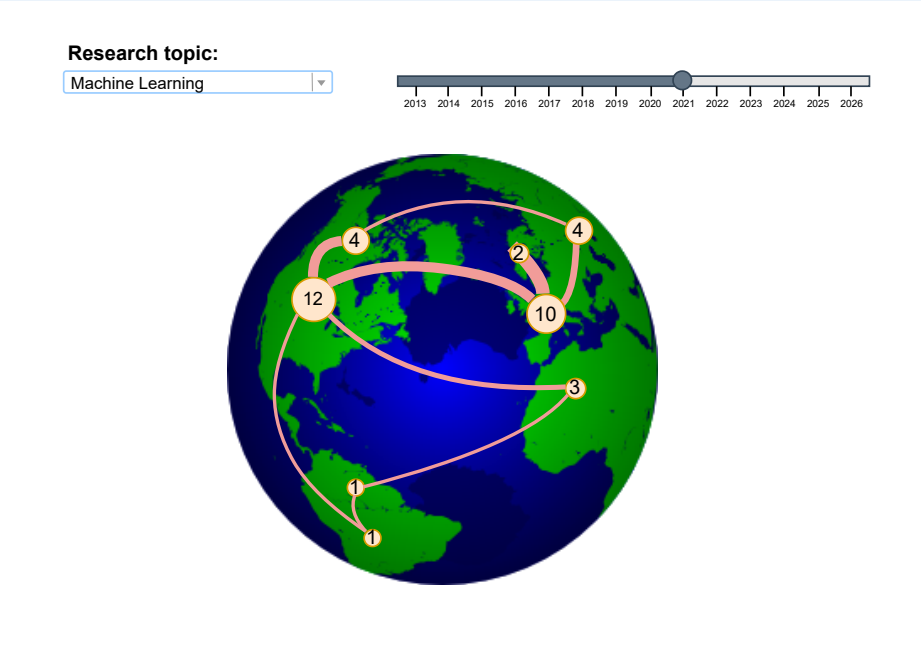
- Is the global academic network becoming more integrated, or are isolated regional clusters forming over time?
- Which countries and cities have transitioned from knowledge consumers to major exporters of research?
- Which university-to-university or country-to-country pairs have developed the strongest reciprocal citation links?

Visualizations & Sketches

1 · Interactive 3D citation globe (Core / MVP)

A rotatable 3D globe with institution nodes and directed citation arc overlays. The user scrubs through years with a timeline slider; arcs animate to show how citation flows shift geographically. Node size encodes citation volume. The globe replaces a flat map, giving an accurate sense of geographic distance and allowing the user to rotate to any region of interest.

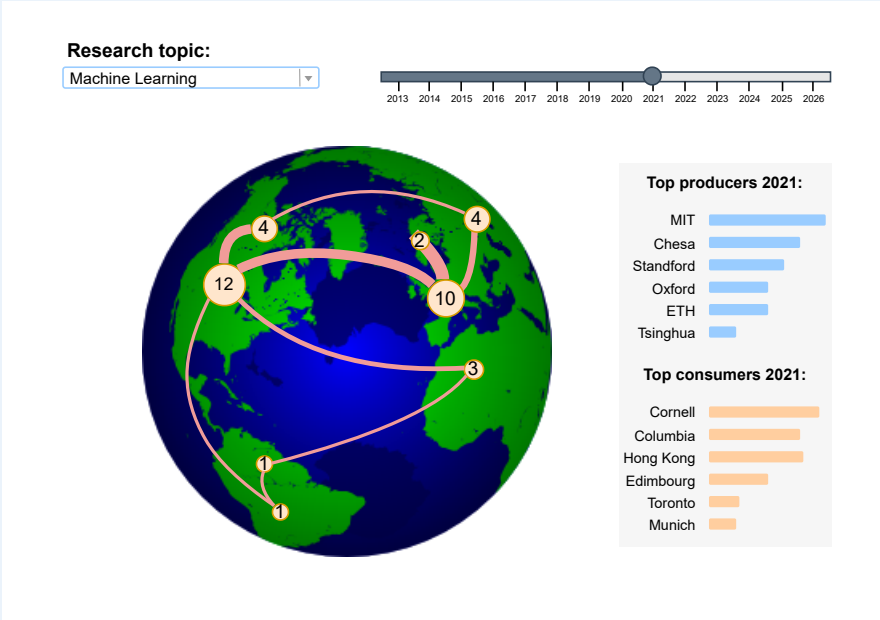
Sketch-1 "**Interactive 3D citation globe**"



2 · Top-5 knowledge producers & consumers leaderboard (Core / MVP)

A live-updating panel showing the top 5 knowledge-producing institutions (most citations given to others) and top 5 knowledge-consuming institutions (most citations received from others) for the selected year. Updates in sync with the year slider.

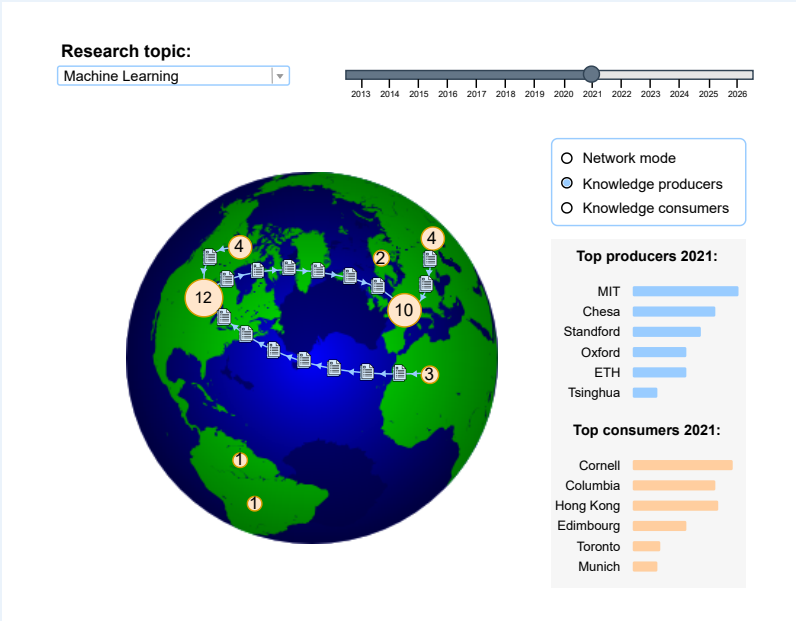
Sketch-2 **"Top-5 knowledge producers & consumers leaderboard"**



3 · Animated citation-volume icons on top producers & consumers (Optional)

Small letter icons (each representing 10,000 citations) radiate outward from the top-5 knowledge producers on the globe and flow inward toward the consumers. The speed of the animation is proportional to the actual citation volume, heavy citation flows animate faster, making the asymmetry between hubs visible.

Sketch-3 **"Animated citation-volume icons"**



4 · Topic / field filter (Optional)

A dropdown or search field lets the user restrict the globe and leaderboard to papers within a specific research topic (e.g. Neuroscience, Machine Learning, LHC, …). The full graph recomputes for the selected topic and year combination, revealing field-specific geographic patterns that are invisible in the aggregate view.

5 · Additional optional features

- Play back the full 2005–2025 period as an auto-advancing animation, showing the rise and fall of regional hubs as a short film.
- Click any two nodes to draw a dedicated arc showing the reciprocal citation count between those two institutions across all years.
- Run a community-detection algorithm (Louvain) on the citation graph and color nodes by detected cluster, revealing citation archipelagos that primarily cite within their own network.
- A small area chart below the globe showing total cross-border edge count by year; scrubbing it also moves the globe to that year.

Tools & Lectures

Visualization	Tools / Libraries	Relevant Lectures
3D citation globe	Three.js / Globe.gl, D3.js, OpenAlex API, Nominatim	Lec: D3.js (9 Mar) · Interactions (16 Mar) · Practical Maps (20 Apr) · Maps (20 Apr)
Leaderboard panel	D3.js (bar chart), React / Svelte state	Lec: D3.js (9 Mar) · Interactions (16 Mar) · Tabular data (11 May)
Letter-icon animation	Three.js sprites, CSS @keyframes	Lec: Interactions (16 Mar) · More interactive D3 (16 Mar) · Perception & colors (23 Mar)
Topic / field filter	OpenAlex topic API, frontend dropdown	Lec: Data (9 Mar) · JavaScript Pt 1 & 2 (23 Feb, 2 Mar)
Community detection overlay	NetworkX (Python, Louvain), D3 color scales	Lec: Graphs (4 May) · Perception & colors (23 Mar) · Marks & Channels (23 Mar)
Network density timeline	D3.js area chart	Lec: Interactions / Time-series (16 Mar) · Storytelling (18 May)

MVP vs. Optional Features

CORE — Minimum Viable Product

- Rotatable 3D globe with institution nodes scaled by citation volume
- Directed arc overlays between institutions for a selected year
- Year scrubber / slider to navigate 2005–2025
- Top-5 knowledge producers & consumers leaderboard, synced to the year slider
- Data pipeline: OpenAlex fetch → Nominatim geocoding → graph construction

OPTIONAL — Can be dropped without affecting core meaning

- Animated letter icons (1 icon = 10k citations) radiating from top producers, flowing into top consumers, at speed proportional to citation volume
- Dropdown to restrict the globe to papers in a specific research domain
- Auto-advancing animation across all years
- Click two nodes to show their reciprocal citation history
- Regional cluster coloring via Louvain community detection
- Network density timeline chart linked to the year scrubber